

BROYDEN'S METHOD

Lecture 17: Presented by Jan van Vuuren



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Broyden's update formula (p. 83)

Broyden's method is perhaps the simplest quasi-Newton method

Its working is based on a rank-one correction term of the form $\Delta^k = \mathbf{u}^k (\mathbf{v}^k)^T$ in the general quasi-Newton update formula $\mathbf{B}^{k+1} \leftarrow \mathbf{B}^k + \Delta^k$

Check that the matrix $\mathbf{u}^k (\mathbf{v}^k)^T$ has rank 1 for any n -vectors $\mathbf{u}^k$ and $\mathbf{v}^k$

If the update formula

$$\mathbf{B}^{k+1} = \mathbf{B}^k + \mathbf{u}^k (\mathbf{v}^k)^T$$

is substituted into the secant condition

$$\mathbf{B}^{k+1} \mathbf{s}^k = \mathbf{y}^k,$$

it follows that

$$(\mathbf{B}^k + \mathbf{u}^k (\mathbf{v}^k)^T) \mathbf{s}^k = \mathbf{y}^k$$

or, equivalently, that

$$\mathbf{B}^k \mathbf{s}^k + \mathbf{u}^k [(\mathbf{v}^k)^T \mathbf{s}^k] = \mathbf{y}^k$$

Since the term in square brackets is a scalar, it follows that

$$\mathbf{u}^k = \frac{\mathbf{y}^k - \mathbf{B}^k \mathbf{s}^k}{(\mathbf{v}^k)^T \mathbf{s}^k}$$

Broyden furthermore proposed taking $\mathbf{v}^k = \mathbf{s}^k$

Broyden's update formula (p. 83)

Broyden's update rule is given by

$$\mathbf{B}^{k+1} = \mathbf{B}^k + \frac{\mathbf{y}^k - \mathbf{B}^k \mathbf{s}^k}{(\mathbf{s}^k)^T \mathbf{s}^k} (\mathbf{s}^k)^T$$

Broyden, in fact, suggested computing the search direction $\mathbf{p}^k$ directly by making use of the Sherman-Morrison formula

$$\mathbf{H}^{k+1} = \mathbf{H}^k + \frac{(\mathbf{s}^k - \mathbf{H}^k \mathbf{y}^k)(\mathbf{s}^k)^T \mathbf{H}^k}{(\mathbf{s}^k)^T \mathbf{H}^k \mathbf{y}^k}$$

in Theorem 1.4 instead of solving the system of linear equations in Step 3 of Algorithm 3.10 by Gauss-Jordan elimination when approximating the Hessian matrix $\nabla^2 f(\mathbf{x}^k)$

That is, approximating the inverse Hessian matrix $[\nabla^2 f(\mathbf{x}^k)]^{-1}$ rather than the Hessian itself

Broyden's method in pseudo-code (p.83)

Algorithm 3.11 (Broyden's method)

Input: An instance of a multivariate differentiable function $f : \mathbb{R}^n \mapsto \mathbb{R}$, an initial estimate $\mathbf{x}^0$ as to an approximate local optimum of f , an initial Hessian estimate $\mathbf{B}^0$ or an initial inverse Hessian estimate $\mathbf{H}^0$, as well as a search termination criterion

Output: An approximation of a local optimum of f close to $\mathbf{x}^0$ satisfying the termination criterion

Step 1 $k \leftarrow 0$

Step 2 If the search termination criterion is met, then output $\mathbf{x}^k$ and stop

Step 3 Compute the search direction $\mathbf{p}^k$ by Gauss-Jordan elimination to solve the system of linear equations $\mathbf{B}^k \mathbf{p}^k = -\nabla f(\mathbf{x}^k)$, or compute it directly as $\mathbf{p}^k = -\mathbf{H}^k \nabla f(\mathbf{x}^k)$

Step 4 Compute an appropriate step length α^k in the search direction $\mathbf{p}^k$ by performing an exact line search or a backtracking Armijo-Goldstein line search in respect of the function $f(\mathbf{x}^k + \alpha^k \mathbf{p}^k)$

Step 5 Compute the next search estimate as $\mathbf{x}^{k+1} \leftarrow \mathbf{x}^k + \alpha^k \mathbf{p}^k$

Step 6 $\mathbf{s}^k \leftarrow \mathbf{x}^{k+1} - \mathbf{x}^k$

Step 7 $\mathbf{y}^k \leftarrow \nabla f(\mathbf{x}^{k+1}) - \nabla f(\mathbf{x}^k)$

Step 8 $\mathbf{B}^{k+1} \leftarrow \mathbf{B}^k + \frac{\mathbf{y}^k - \mathbf{B}^k \mathbf{s}^k}{(\mathbf{s}^k)^T \mathbf{s}^k} (\mathbf{s}^k)^T$ or $\mathbf{H}^{k+1} \leftarrow \mathbf{H}^k + \frac{(\mathbf{s}^k - \mathbf{H}^k \mathbf{y}^k)(\mathbf{s}^k)^T \mathbf{H}^k}{(\mathbf{s}^k)^T \mathbf{H}^k \mathbf{y}^k}$

Step 9 $k \leftarrow k + 1$ and return to Step 2

Convergence of Broyden's method (p. 84)

Theorem 3.6 (Convergence rate of Broyden's method)

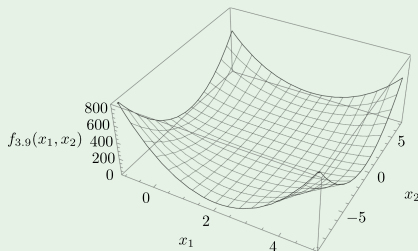
Let $f : \mathbb{R}^n \mapsto \mathbb{R}$ be a twice continuously differentiable, Lipschitz-continuous function with a local optimum $\mathbf{x}^*$ at which the Hessian matrix $\nabla^2 f(\mathbf{x}^*)$ is invertible. Then there exist positive real numbers δ and ε such that the sequence $\mathbf{x}^0, \mathbf{x}^1, \mathbf{x}^2 \dots$ of search points generated by Algorithm 3.11 converges super-linearly to $\mathbf{x}^*$ whenever $\|\mathbf{x}^0 - \mathbf{x}^*\|_p < \delta$ and such that $\|\mathbf{B}^0 - \nabla^2 f(\mathbf{x}^*)\| < \varepsilon$, where $\|\cdot\|$ is an appropriate matrix norm.

The theorem essentially states that Broyden's method will converge to a local optimum $\mathbf{x}^*$ of a twice continuously differentiable function $f(\mathbf{x})$

- if the initial search point $\mathbf{x}^0$ is chosen sufficiently close to $\mathbf{x}^*$, and
- if the initial estimate $\mathbf{B}^0$ of the Hessian matrix $\nabla^2 f(\mathbf{x})$ is chosen sufficiently close to $\nabla^2 f(\mathbf{x}^*)$

Example 3.15 (pp. 84–87)

$$f_{3.9}(x_1, x_2) = 17 - 32x_1 + 24x_1^2 - 8x_1^3 + x_1^4 + 2x_2 + 5x_2^2 - 4x_1x_2^2 + x_1^2x_2^2$$



We apply Algorithm 3.11 (equipped with a Armijo-Goldstein line search with parameters $a = 0.4$ and $b = 0.9$) to minimise $f_{3.9}$, starting out from $\mathbf{x}^0 = [-1 \ 1]^T$

We are given an initial symmetric, positive definite Hessian estimate

$$\mathbf{B}^0 = \begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix}$$

We work accurately to six decimal places and terminate the iterative process once the 2-norm $\|\nabla f_{3.9}(\mathbf{x}^k)\|_2$ drops below 0.05 for the first time.

Example 3.15 (pp. 84–87)

Our first step is to compute the gradient vector of $f_{3.9}$ as

$$\nabla f_{3.9}(x_1, x_2) = \begin{bmatrix} 2(x_1 - 2)(8 - 8x_1 + 2x_1^2 + x_2^2) \\ 2 + 2(5 - 4x_1 + x_1^2)x_2 \end{bmatrix}$$

During Iteration $k = 0$, the gradient vector is $\nabla f_{3.9}(\mathbf{x}^0) = [-114 \ 22]^T$

We compute the search direction by solving the system

$$\begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix} \begin{bmatrix} p_1^0 \\ p_2^0 \end{bmatrix} = \begin{bmatrix} 114 \\ -22 \end{bmatrix}$$

The first search direction $\mathbf{p}^0 = [\frac{46}{5} \ \frac{37}{10}]^T = [9.2 \ 3.7]^T$ is obtained

A backtracking Armijo-Goldstein line search with parameters $a = 0.4$ and $b = 0.9$ yields the step length $\alpha^0 = 0.205891$

As a result, $\mathbf{s}^0 = \alpha^0 \mathbf{p}^0 \approx [1.894198 \ 0.761797]^T$

Therefore, $\mathbf{x}^1 = \mathbf{x}^0 + \alpha^0 \mathbf{p}^0 \approx [0.894198 \ 1.761797]^T$

Moreover, $\mathbf{y}^0 = \nabla f_{3.9}(\mathbf{x}^1) - \nabla f_{3.9}(\mathbf{x}^0) \approx [101.726656 \ -12.167764]^T$

Example 3.15 (pp. 84–87)

The Hessian estimate may consequently be updated to

$$\begin{aligned} B^1 &= B^0 + \frac{\mathbf{y}^0 - B^0 \mathbf{s}^0}{(\mathbf{s}^0)^T \mathbf{s}^0} (\mathbf{s}^0)^T \\ &= \begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix} + \frac{1}{(1.894\,198)^2 + (0.761\,797)^2} \\ &\quad \times \left(\begin{bmatrix} 101.726\,656 \\ -12.167\,764 \end{bmatrix} - \begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix} \begin{bmatrix} 1.894\,198 \\ 0.761\,797 \end{bmatrix} \right) [1.894\,198 \quad 0.761\,797] \\ &\approx \begin{bmatrix} 49.561\,217 & 10.301\,794 \\ -7.470\,986 & 2.604\,060 \end{bmatrix}. \end{aligned}$$

Since

$$\|\nabla f_{3.9}(\mathbf{x}^0)\|_2 \approx 116.103\,402 > 0.05,$$

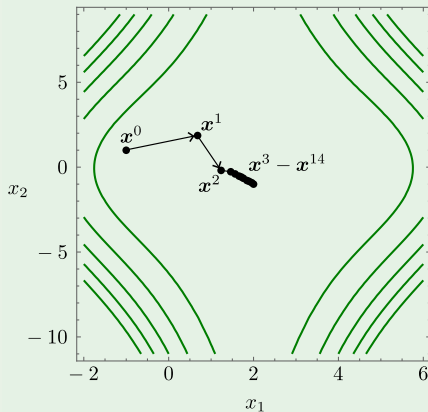
Algorithm 3.11 does not terminate immediately after the first iteration ...

Example 3.15 (pp. 84–87)

k	$\mathbf{x}^k$	$\nabla f(\mathbf{x}^k)$	$\mathbf{B}^k$	$\mathbf{p}^k$	α^k	$\mathbf{s}^k$	$\mathbf{y}^k$	$\ \nabla f(\mathbf{x}^k)\ _2$
0	$\begin{bmatrix} -1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} -114 \\ 22 \end{bmatrix}$	$\begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix}$	$\begin{bmatrix} 9.2 \\ 3.7 \end{bmatrix}$	0.206	$\begin{bmatrix} 1.894 \\ 0.762 \end{bmatrix}$	$\begin{bmatrix} 101.727 \\ -12.168 \end{bmatrix}$	116.103
1	$\begin{bmatrix} 0.894 \\ 1.762 \end{bmatrix}$	$\begin{bmatrix} -12.273 \\ 9.832 \end{bmatrix}$	$\begin{bmatrix} 49.561 & 10.302 \\ -7.470 & 2.604 \end{bmatrix}$	$\begin{bmatrix} 0.647 \\ -1.920 \end{bmatrix}$	1	$\begin{bmatrix} 0.647 \\ -1.920 \end{bmatrix}$	$\begin{bmatrix} 11.863 \\ -8.216 \end{bmatrix}$	15.726
2	$\begin{bmatrix} 1.541 \\ -0.158 \end{bmatrix}$	$\begin{bmatrix} -0.410 \\ 1.617 \end{bmatrix}$	$\begin{bmatrix} 49.497 & 10.494 \\ -7.216 & 1.848 \end{bmatrix}$	$\begin{bmatrix} 0.106 \\ -0.461 \end{bmatrix}$	1	$\begin{bmatrix} 0.106 \\ -0.461 \end{bmatrix}$	$\begin{bmatrix} -0.037 \\ -1.009 \end{bmatrix}$	1.667
3	$\begin{bmatrix} 1.647 \\ -0.619 \end{bmatrix}$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
12	$\begin{bmatrix} 1.952 \\ -0.994 \end{bmatrix}$	$\begin{bmatrix} -0.096 \\ 0.007 \end{bmatrix}$	$\begin{bmatrix} -0.308 & -5.847 \\ 0.738 & 3.129 \end{bmatrix}$	$\begin{bmatrix} 0.077 \\ -0.020 \end{bmatrix}$	0.729	$\begin{bmatrix} 0.056 \\ -0.015 \end{bmatrix}$	$\begin{bmatrix} 0.112 \\ -0.025 \end{bmatrix}$	0.096
13	$\begin{bmatrix} 2.008 \\ -1.009 \end{bmatrix}$	$\begin{bmatrix} 0.016 \\ -0.018 \end{bmatrix}$	$\begin{bmatrix} 0.386 & -6.031 \\ 0.405 & 3.217 \end{bmatrix}$	$\begin{bmatrix} 0.016 \\ 0.004 \end{bmatrix}$	1.1×10^{-10}	$\begin{bmatrix} 1.7 \times 10^{-12} \\ 3.8 \times 10^{-13} \end{bmatrix}$	$\begin{bmatrix} 3.4 \times 10^{-12} \\ 7.1 \times 10^{-13} \end{bmatrix}$	0.024
14	$\begin{bmatrix} 2.008 \\ -1.009 \end{bmatrix}$	—	—	—	—	—	—	—

Example 3.15 (pp. 84–87)

Broyden's method returns $\mathbf{x}^{14} = [2.007\ 679 \quad -1.008\ 922]^T$ as approximate minimum of $f_{3.9}$; Compare this to the true minimum is $\mathbf{x}^* = [2 \quad -1]^T$



Disadvantages of Broyden's method (p. 88)

Two disadvantages of Broyden's update formula are that it preserves neither symmetry nor positive definiteness of the Hessian estimates $\mathbf{B}^0, \mathbf{B}^1, \mathbf{B}^2, \dots$

That is, even if $\mathbf{B}^k$ is a symmetric matrix, it does not necessarily hold that $\mathbf{B}^{k+1}$ is also symmetric

This would seem to be a serious shortcoming, given that the matrix $\mathbf{B}^k$ is meant to estimate the Hessian matrix $\nabla^2 f(\mathbf{x}^k)$, which is indeed symmetric for any twice continuously differentiable function $f(\mathbf{x}^k)$

Similarly, even if $\mathbf{B}^k$ is a positive definite [negative definite, resp.] matrix, it does not necessarily hold that $\mathbf{B}^{k+1}$ is also positive definite [negative definite, resp.]

This means that an ascent/descent guaranteeing correction similar to that in Step 5 of Algorithm 3.7 might have to be implemented repeatedly during execution of the algorithm when positive definiteness [negative definiteness, resp.] is lost, as appropriate